Process FMEA — Part 34521-A (Hydraulic Actuator Housing)

Rev B · effective 2025-07-15. Team: S. Okafor (QE, lead), R. Bertolino (Process Eng), D. Hoang (Cell Lead). Methodology: AIAG-VDA FMEA Handbook 1st ed., 7-step approach. Cross-reference: control plan `QMS-CP-34521A-RevC`.


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--- METADATA ---
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domain: quality
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owner: S. Okafor, Quality Engineering — Cleveland
part_numbers: [34521-A]
operation_numbers: [OP20, OP30, OP40]
machine_ids: [NLX-08, NLX-09]
controller_models: [Fanuc 0i-TF]
fault_codes: []
materials: [MAT-4140]
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```


1. Scope and assumptions

This PFMEA covers turning and inspection operations OP20–OP40 for Part `34521-A` as run in the Auto-Turn-1 cell on machines `NLX-08` and `NLX-09`. OP10 (stock saw cutoff on `BSAW-03`) is covered in the band-saw cell PFMEA. Severity, Occurrence, and Detection (Action Priority not shown in this excerpt) follow the AIAG-VDA 10-point scales.

2. Failure mode table

ID	Operation / function	Failure mode	Effect on customer	S	Cause(s)	O	Current process control	D
FM-001	OP20 — Rough turn OD	OD undersize	Stock allowance insufficient for OP30; scrap.	5	Tool offset error; insert wear past limit.	3	Micrometer check every 25 pcs; offset compensation table.	4

FM-002	OP20 — Rough turn OD	OD oversize	Rework at OP30; possible chatter at finish.	4	Tool offset error.	3	Micrometer check every 25 pcs.	4
FM-007	**OP30 — Finish turn + bore**	**Bore diameter out of tolerance (under or over Ø 38.000 +0.025 / −0.000 mm)**	**Loss of seal on hydraulic actuator; field failure; safety-related warranty event.**	**8**	(a) Boring-bar insert wear past flank-wear limit VBmax = 0.20 mm. (b) Thermal drift after warm-up cycle interrupted. (c) Work-offset entry error after tool change. (d) Coolant concentration drift causing built-up edge.	5	Bore gauge check every 10 pcs per `QMS-CP-34521A-RevC`; CMM 100% at OP40; in-process insert change at 200 pcs.	3
FM-008	OP30 — Finish turn + bore	Bore surface finish above Ra 0.8 µm	Seal life reduced; warranty exposure.	6	Insert wear; incorrect feed override; coolant concentration low.	4	Profilometer check per setup; coolant refractometer check daily.	4
FM-009	OP30 — Finish turn + bore	Bore concentricity to OD > 0.05 mm TIR	Side load on seal; reduced life.	6	Worn collet or chuck jaws; coolant chip in fixture.	2	CMM 100% at OP40; quarterly fixture wear check.	3
FM-014	OP40 — CMM inspection	False accept (gauge R&R drift)	Out-of-tolerance pieces ship to Vortex Motors.	9	CMM probe wear; thermal soak skipped.	1	Annual gauge R&R; daily probe qualification.	3

3. Failure-mode → reaction-plan linkage

The following failure modes have explicit reaction plans in the control plan (`QMS-CP-34521A-RevC`). Use this map when investigating a rejection on the floor.

If the rejection is …	… the failure mode is …	… the reaction plan is in …
Bore diameter out of tolerance	**FM-007**	Control plan OP30 row + §3 (full reaction plan, including IATF §10.2.1 timing)

Bore surface finish above Ra 0.8 µm	FM-008	Control plan OP30 row (replace insert, verify, resume)
Bore concentricity above 0.05 mm TIR	FM-009	Control plan OP40 row (MRB; investigate fixture)

> **Hint for cross-source agent queries:** A floor query of the form "we have a bore-diameter rejection on Part 34521-A — what does the control plan say and which FMEA mode does this correspond to?" resolves to FM-007 + control plan §3, then routes to `PLM-WI-34521A-OP30-RevD` for the correct CNC turning parameters to verify on restart.

4. Review history

Revision	Date	Change	Trigger
Rev A	2024-01-20	Initial issue.	New part launch.
Rev B	2025-07-15	Severity of FM-007 raised from 7 to 8 after field warranty data; detection improved to 3 with bore-gauge frequency change.	Customer corrective action request from Vortex Motors (CAR-VTX-2025-014).

Document control / sign-off

Role	Name	Signature	Date
Document Owner	S. Okafor, Quality Engineering — Cleveland	(on file)	2025-07-15
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